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CS202 – ADVANCED OO PROGRAMMING

CHAPTER 3

FILE HANDLING IN JAVA

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Agenda

1. What is File Handling in Java?
2. What is a stream?
3. Java File Methods
4. File operations in Java

File Handling in Java

File Handling implies how to read from and write to a file in Java.

➔ Java provides the basic I/O package for reading and writing streams.

➔ java.io package allows to do all input and output tasks in Java

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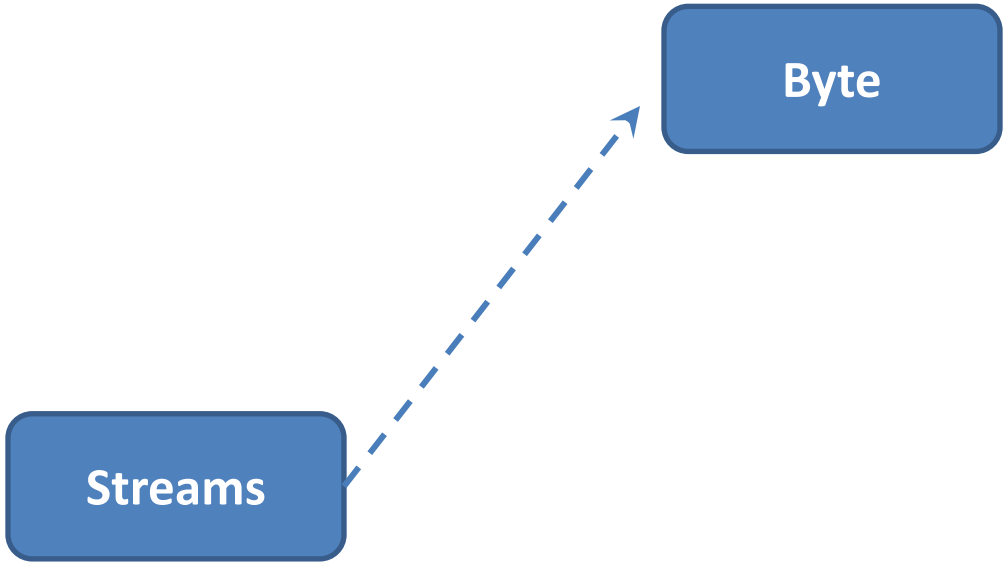
What is a stream?



used to perform input and output of 8-bit bytes
java.io package {video, audio, characters}

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Byte stream example

Classes : **FileInputStream**

FileOutputStream

```
import java.io.*;
public class CopyFile {

  public static void main(String args[]) throws IOException {
     FileInputStream in = null;
     FileOutputStream out = null;

     try {
          in = new FileInputStream("input.txt");
          out = new FileOutputStream("output.txt");
          int c;
          while ((c = in.read()) != -1) {
               out.write(c);}
      }finally {
         if (in != null)
               in.close();
         if (out != null)
               out.close();  }

   }
}
```

Assumption: *"input.txt"* File contains the following sentence

This is test for copy file.

The compilation and execution of the program ***CopyFile*** creates the *"output.txt"* with the same content as the first file.

Commands

$javac CopyFile.java

$java CopyFile

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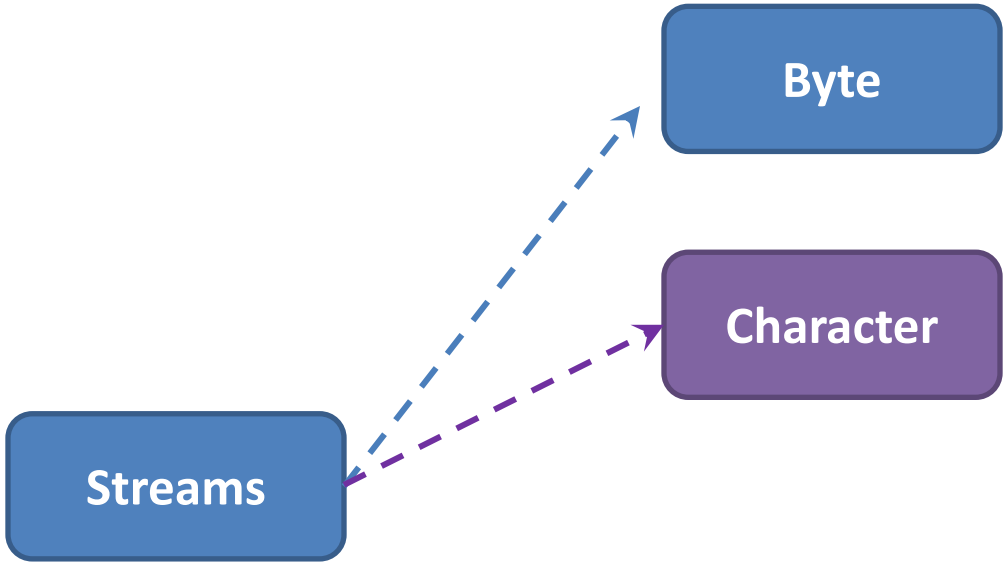
What is a stream?



used to perform input and output of 8-bit bytes
java.io package {video, audio, characters}

used to perform input and output of 16-bit unicode
java.io package

Reader	Writer
BufferedReader	BufferedWriter
CharacterArrayReader	CharacterArrayWriter
StringReader	StringWriter
FileReader	FileWriter
InputStreamReader	InputStreamWriter
FileReader	FileWriter

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Character stream example

Abstract classes : **Reader** → **FileReader** → **FileInputStream**

Writer → **FileWriter** → **FileOutputStream**

```
import java.io.*;
public class CopyFile {
     public static void main(String args[]) throws IOException {
          FileReader in = null;
          FileWriter out = null;
          try {
               in = new FileReader("input.txt");
               out = new FileWriter("output.txt");
               int c;
               while ((c = in. read()) != -1)
                    out. write(c); }
          finally {
                if (in != null)
                     in.close();
                if (out != null)
                     out.close(); }
     }
 }
```

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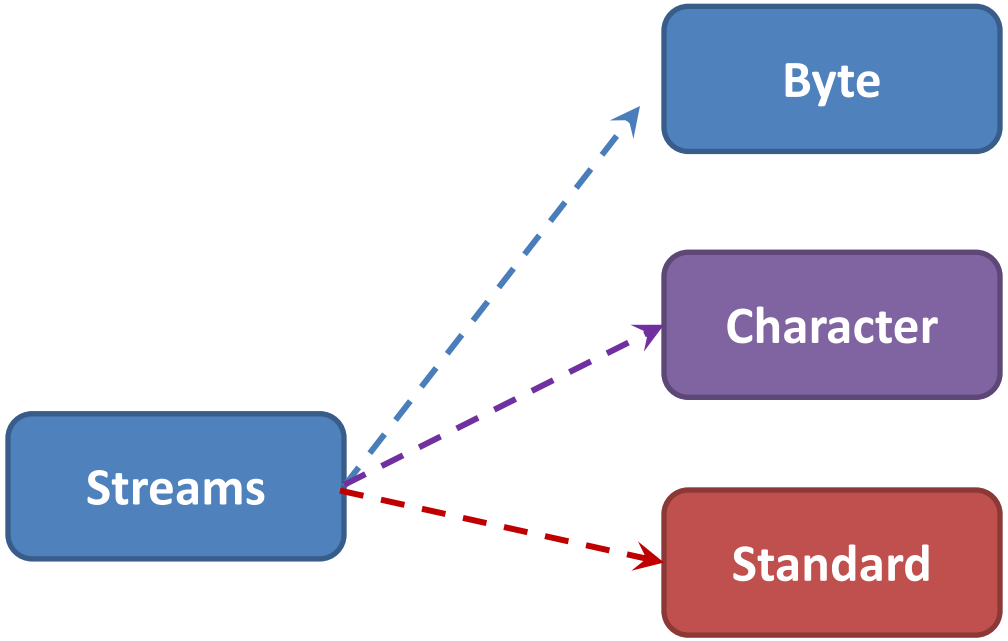
What is a stream?



used to perform input and output of 8-bit bytes
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can take input from a keyboard and then produce an output on the computer screen
java.io package {System.in, System.out, System.err}

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What is a stream?

Standard

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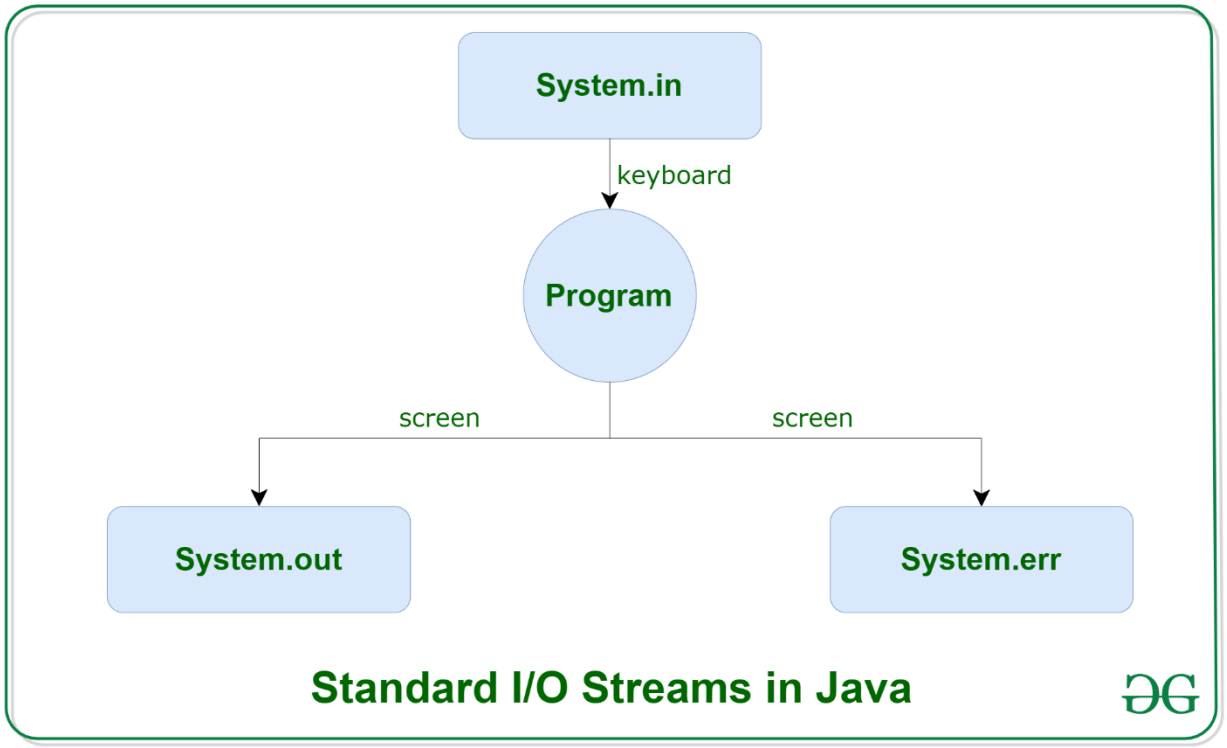

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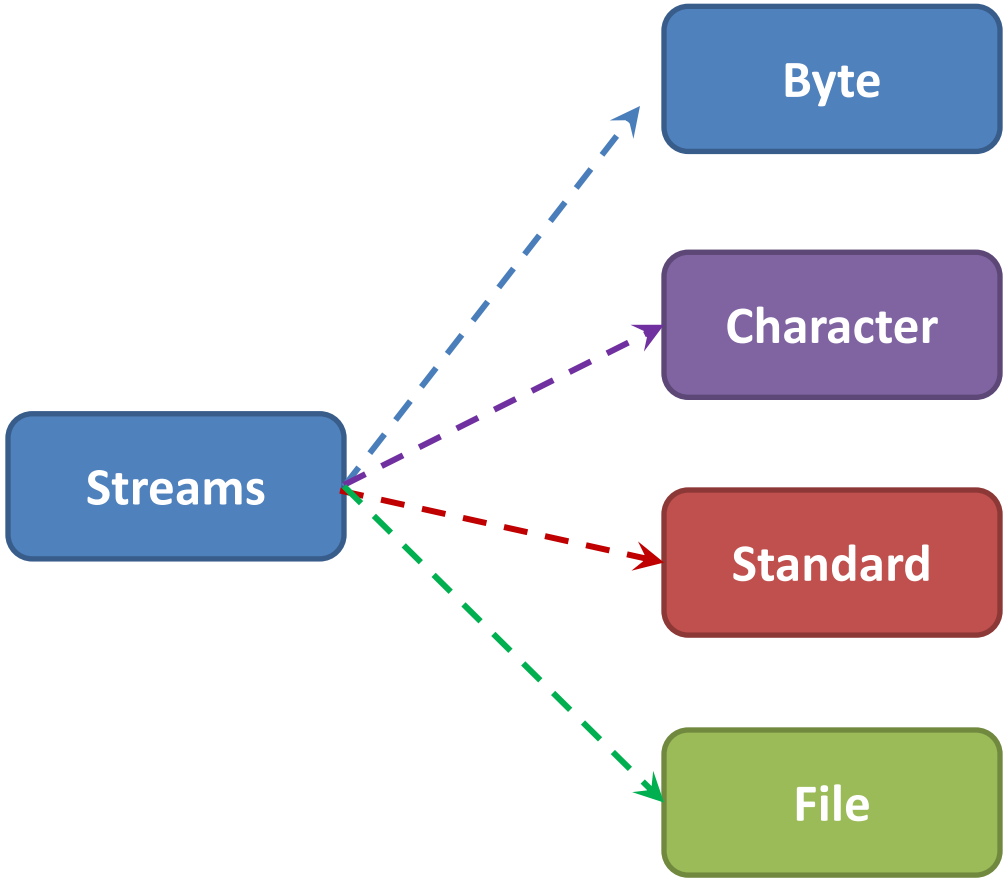
Standard stream example

```
import java.io.*;
public class ReadConsole {
   public static void main(String args[]) throws IOException {
         InputStreamReader cin = null;
         try {
              cin = new InputStreamReader(System.in);
              System.out.println("Enter characters, 'q' to
                     quit.");
              char c;
              do {
                      c = (char)cin.read();
                      System.out.print(c);
              } while(c != 'q');
         }finally {
             if (cin!= null)
                cin.close();}
      }
   }
```

What is a stream?



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can take input from a keyboard and then produce an output on the computer screen
java.io package {System.in, System.out, System.err}

This stream is used for reading data from a file and writing information to it.
java.io package {FileInputStream, FileOutputStream}

Java File methods

Method	Type	Description
canRead()	Boolean	Tests whether the file is readable or not
canWrite()	boolean	Tests whether the file is writable or not

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Java File methods

Method	Type	Description
canRead()	Boolean	Tests whether the file is readable or not
canWrite()	boolean	Tests whether the file is writable or not
createNewFile()	boolean	Creates an empty file
delete()	boolean	Deletes a file

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canWrite()	boolean	Tests whether the file is writable or not
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getName()	String	Returns the name of the file

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getName()	String	Returns the name of the file
getAbsolutePath()	String	Returns the absolute pathname of the file

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exists()	boolean	Tests whether the file exists
getName()	String	Returns the name of the file
getAbsolutePath()	String	Returns the absolute pathname of the file
length()	Long	Returns the size of the file in bytes

Java File methods

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canWrite()	boolean	Tests whether the file is writable or not
createNewFile()	boolean	Creates an empty file
delete()	boolean	Deletes a file
exists()	boolean	Tests whether the file exists
getName()	String	Returns the name of the file
getAbsolutePath()	String	Returns the absolute pathname of the file
length()	Long	Returns the size of the file in bytes
list()	String[]	Returns an array of the files in the directory

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Java File methods

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canRead()	Boolean	Tests whether the file is readable or not
canWrite()	boolean	Tests whether the file is writable or not
createNewFile()	boolean	Creates an empty file
delete()	boolean	Deletes a file
exists()	boolean	Tests whether the file exists
getName()	String	Returns the name of the file
getAbsolutePath()	String	Returns the absolute pathname of the file
length()	Long	Returns the size of the file in bytes
list()	String[]	Returns an array of the files in the directory
mkdir()	Boolean	Creates a directory

File methods

boolean canExecute() : Tests whether the application can execute the file denoted by this abstract pathname.
int compareTo(File pathname) : Compares two abstract pathnames lexicographically.
static File createTempFile(String prefix, String suffix) : Creates an empty file in the default temporary-file directory.
boolean equals(Object obj) : Tests this abstract pathname for equality with the given object.
long getFreeSpace() : Returns the number of unallocated bytes in the partition .
String getParent() : Returns the pathname string of this abstract pathname's parent.
File getParentFile() : Returns the abstract pathname of this abstract pathname's parent.
String getPath() : Converts this abstract pathname into a pathname string.
boolean isDirectory() : Tests whether the file denoted by this pathname is a directory.
boolean isFile() : Tests whether the file denoted by this abstract pathname is a normal file.
boolean isHidden() : Tests whether the file named by this abstract pathname is a hidden file.
String[ ] list() : Returns an array of strings naming the files and directories in the directory .
File[ ] listFiles() : Returns an array of abstract pathnames denoting the files in the directory.
boolean mkdir() : Creates the directory named by this abstract pathname**.**
boolean renameTo(File dest) : Renames the file denoted by this abstract pathname.
boolean setExecutable(boolean executable) : A convenience method to set the owner's execute permission.
boolean setReadable(boolean readable) : A convenience method to set the owner's read permission.
boolean setReadable(boolean readable, boolean ownerOnly) : Sets the owner's or everybody's read permission.
boolean setReadOnly() : Marks the file or directory named so that only read operations are allowed.
boolean setWritable(boolean writable) : A convenience method to set the owner's write permission.
String toString() : Returns the pathname string of this abstract pathname.
URI toURI() : Constructs a file URI that represents this abstract pathname.

File operations in Java

Example 1 – Create a file

```
import java.io.File;  //import File class
import java.io.IOException; //import IOException class to handle errors

public class CreateFile {
        public static void main(String[] args) {
            try {
                File myObj = new File("filename.txt");
                if (myObj.createNewFile())
                        System.out.println("File created: "+ myObj.getName());

                else
                        System.out.println("File already exists.");
            }catch (IOException e) {
                        System.err.println("An error occurred.");
                        e.printStackTrace();}
        }
}
```

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File operations in Java

01 Creation of a file

02 Getting information about a file

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Example 2 – Get file properties

```
import java.io.File;
class fileProperties
{
    public static void main(String[] args) { /*accept file name or directory name through command
                                                line args*/
        String fname =args[0]; //pass the filename or directory name to File object
        File f = new File(fname);
        System.out.println("File name : "+f.getName());
        System.out.println("Path : "+f.getPath());
        System.out.println("Absolute path : " +f.getAbsolutePath());
        System.out.println("Parent : "+f.getParent());
        System.out.println("Exists : "+f.exists());
        if(f.exists())
        {
            System.out.println("Is writeable : "+f.canWrite());
            System.out.println("Is readable :  "+f.canRead());
            System.out.println("Is a directory : "+f.isDirectory());
            System.out.println("File Size in bytes : "+f.length());
        }
    }
```

File operations in Java

01 Creation of a file

02 Getting information about a file

03 Writing in a file

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Example 3 – write to file

```
import java.io.FileWriter; //import the FileWriter class
import java.io.IOException; //import the IOException class to handle errors

public class WriteToFile {
    public static void main(String[] args) {
        try {
                FileWriter myWriter = new FileWriter("filename.txt");
                myWriter.write("Files in Java might be tricky, but it is fun enough!");
                myWriter.close();
                System.out.println("Successfully wrote to the file.");
        } catch (IOException e) {
                System.err.println("An error occurred.");
                e.printStackTrace();}
    }
}
```

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File operations in Java

01 Creation of a file

02 Getting information about a file

03 Writing in a file

04 Reading from a file

Example 4 – read from file

```
import java.io.File;     // Import the File class
import java.io.FileNotFoundException;    // Import this class to handle errors
import java.util.Scanner; // Import the Scanner class to read text files

public class ReadFile {
    public static void main(String[] args) {
        try {
                File myObj = new File("filename.txt");
                Scanner myReader = new Scanner(myObj);
                while (myReader.hasNextLine()) {
                        String data = myReader.nextLine();
                        System.out.println(data); }
                myReader.close();
        } catch (FileNotFoundException e) {
            System.err.println("An error occurred.");
            e.printStackTrace();}
    }
}
```

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